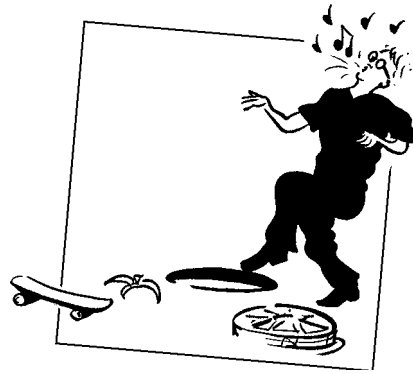


CHAPTER 26

derivatives **** ups



The aim of this Chapter...

... is to warn the reader of the dangers associated with the naive use of derivatives, and to encourage the application of common sense at all times. The chapter is also a nice break from the intense mathematics of preceding chapters.

In this Chapter...

- Orange County
- Proctor and Gamble
- Metallgesellschaft
- Gibson Greetings
- Barings
- LTCM



26.1 INTRODUCTION

Derivatives, in the wrong hands, can be dangerous weapons. They can destroy careers and institutions. In this chapter we take a look of some of the more well-publicized cases of derivatives **** ups. In many cases the details are not in the public domain but where possible I include what is known, and offer some analysis.

26.2 ORANGE COUNTY

Orange County is in California. For the first half of the 1990s the County Treasurer was the aptly named Robert Citron. He was in charge of the County investment fund, a pool of money into which went various taxes of the townsfolk. From 1991 until the beginning of 1994 Citron steadily made money, totaling some \$750 million, by exploiting low interest rates. This was a very good return on investment, representing approximately 400 basis points above US government rates. How was this possible? He had invested the good people's money in leveraged inverse floating rate notes.

A floating rate note is a bond that pays coupons linked to a floating interest rate such as three- or six-month LIBOR. Normally, the coupon rises when rates rise, not so with inverse floaters. Typically inverse floaters have a coupon of the form

$$\max(\alpha r_f - r_L, 0)$$

where r_f is a fixed rate, r_L some LIBOR rate and $\alpha > 1$ a multiplicative factor. As rates rise, the coupons fall but with a floor at zero; the bond holder never has to return money. Citron bought leveraged inverse floaters¹ having coupons of the form

$$\max(\alpha r_f - \beta r_L, 0)$$

with $\beta > 1$. (If $\beta < 1$ it is a deleveraged note.) While rates are low, coupons are high. This was the situation in the early 1990s, and Citron and Orange County benefited. But what if rates rise? These notes have a high degree of gearing.

Citron gambled that rates would stay low.

In mid-1994 US rates rose dramatically, by a total of 3%. The leveraging in the notes kicked in big time and Orange County lost some \$1.6 billion. Figure 26.1 shows US interest rates during the early 1990s. For about 18 months short-term yields had been below 4%, the horizontal line in the figure. But in early 1994 they started a rise that was dramatic in comparison with the previous period of 'stability.' Citron (and others, see below) were 'caught out' by this rise (to the right of the vertical line in the figure).

Orange County didn't have to declare bankruptcy, they were still in the black and money was still pouring into the fund. But as part of the tactics in their damage suit against the brokers of the deal, they declared bankruptcy on 6th December 1994.

It seems that no one had much of a clue how the losses were piling up. There was no frequent marking to market necessary. Had there been, presumably the losses would have been anticipated and someone would have taken action. At the time of the bankruptcy

¹ Actually, he bought a lot of other things as well. He was not a lucky boy, our Bob.

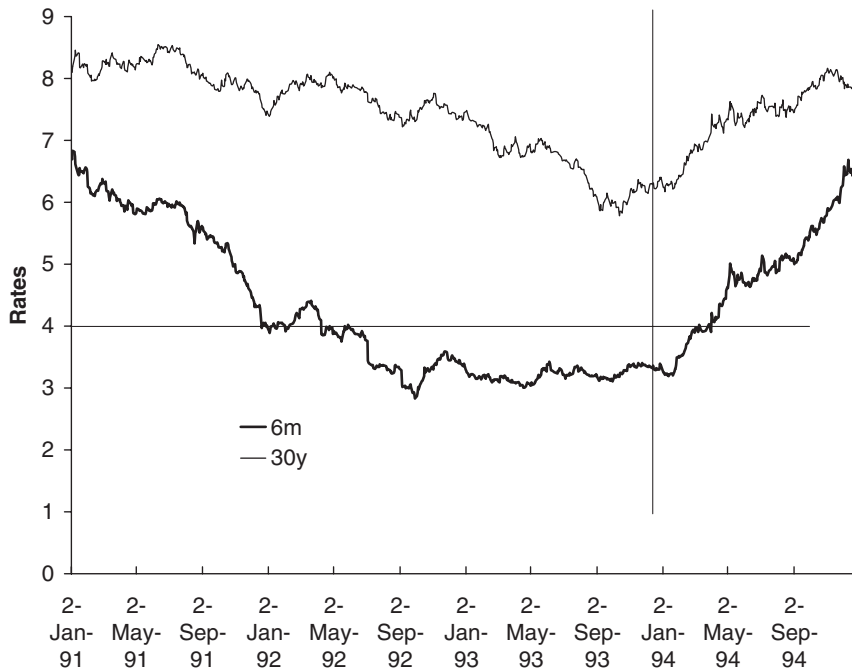


Figure 26.1 US rates in the early 1990s.

filing S&P's rated Orange County as AA and Moody's as Aa1, very high ratings and thought very unlikely to default.

'Leveraged inverse floating rate note' is a very long name for a very simple instrument. The risks should be obvious. After all, how difficult is it to understand $\max(\alpha r_f - \beta r_L, 0)$?

If Orange County lost who gained? The counterparties selling the floaters were various US government housing agencies. Amusingly, the money flowed from the Orange County taxpayers to the US taxpayers everywhere. Hee, hee.

Citron was found guilty of violating state investment laws and was sentenced to one year of community service. During the sentencing phase psychologists found that he had the math skills of a seventh grader and that he was in the lowest 5% of the population in terms of ability to think and reason.

26.3 PROCTOR AND GAMBLE

Proctor and Gamble (P&G) is a major multinational company who manufactures beauty and health care products, food and beverages, and laundry and cleaning products. They have a large exposure to interest rates and to exchange rates. To reduce this exposure they use interest rate and currency swaps.

In late 1993 P&G wanted to enter into a swap from fixed to floating, having the view that rates then low would remain low. A vanilla swap would be fine as long as rates didn't rise, but what if they did? Bankers Trust (BT), the counterparty to the deal, suggested some modifications to the swap that satisfied P&G's concerns.

The deal, struck on 2nd November 1993 was a five-year swap on a notional \$200 million. It contained something a little out of the ordinary, but not outrageously so. P&G had sold BT something like a put on long-term bond prices.

The deal went like this. BT pays P&G a fixed rate of interest on the \$200 million for five years. In return P&G pays BT a fixed rate for the first six months, thereafter a rate defined by

$$r_C - 0.0075 + 0.01 \times \max\left(\frac{98.5}{5.78}Y_5 - P_{30}, 0\right) \quad (26.1)$$

where r_C was the rate on P&G's own corporate bonds, Y_5 the five-year Treasury yield and P_{30} the price of the 30-year Treasury bond. The Treasury yield and price would be known at the time of the first payment, 2nd May 1994, *at which time it would be fixed in the formula*. In other words, the yield and price pertaining on that date would be locked in for the remaining four and a half years.

The best that P&G could achieve would be for rates to stay near the level of November 1993 for just a few more months in which case they would benefit by

$$0.0075 \times \$200 \text{ million} \times 5 = \$7.5 \text{ million.}$$

Not a vast amount in the scheme of things. Five- and 30-year rates had been falling fairly steadily for the whole of the 1990s so far, see Figure 26.1, perhaps they would continue to do so, matching the stability of the short-term yields.

However, if rates were to rise between November and May . . .

Expression (26.1) increases as the five-year yield increases and decreases if the 30-year bond rises in value. But, of course, if the 30-year yield rises the bond price falls and (26.1) increases. Although there is some small exposure to the slope of the yield curve, the dominant effect is due to the level of the yield curve.

In November 1993 the 6.25% coupon bond maturing in August 2023 had a price of about 103.02, corresponding to a yield of approximately 5.97%. The five-year rate was around 4.95%. With those values expression (26.1) was safely the required $r_C - 0.0075$. However, rates rose at the beginning of 1994 and the potential \$7.5 million was not realized, instead P&G lost close to \$200 million.

Subsequently, P&G sued BT on the grounds that they failed to disclose pertinent information. The case was settled out of court.

The following was taken from P&G Corporate News Releases

P&G Settles Derivatives Lawsuit With Bankers Trust
May 9, 1996

CINCINNATI, May 9, 1996—The Procter & Gamble Company today reached an agreement to settle its lawsuit against Bankers Trust. The suit involves two derivative contracts on which Bankers Trust claimed P&G owed approximately \$200 million. Under the terms of the agreement, P&G will absorb \$35 million of the amount in dispute, and Bankers Trust will absorb the rest, or about 83% of the total.

'We are pleased with the settlement and are glad to have this issue resolved,' said John E. Pepper, P&G chairman and chief executive.

It's not difficult to work out the potential losses *a priori* from a shift in the yield curve, and I've done just that in Table 26.1 assuming a parallel shift. P&G start to lose out after about

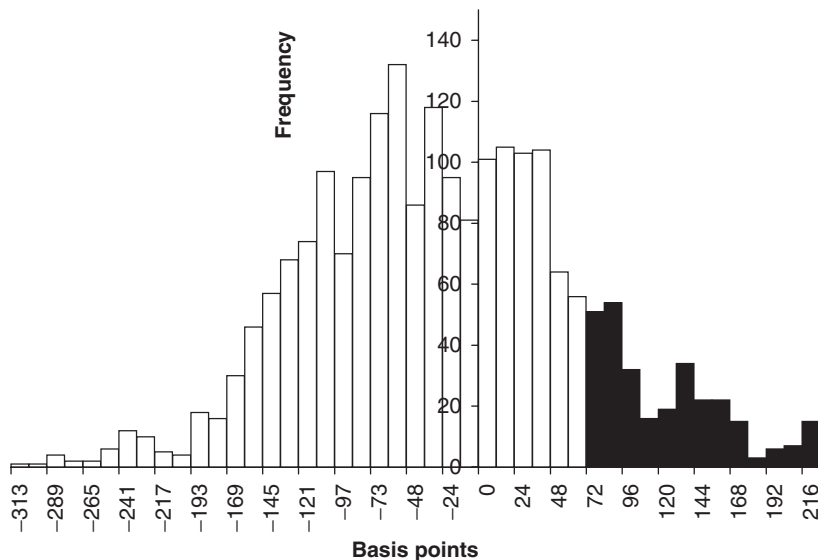
Table 26.1 Effect of parallel shift in yield curve on P&G's losses.

Parallel shift (bps)	0	50	100	150	200
Five-year yield	4.95%	5.45%	5.95%	6.45%	6.95%
Price of 30-year bond	103.02	97.77	93.04	88.74	84.82
30-year yield	5.97%	6.47%	6.97%	7.47%	7.97%
Total loss over 4.5 years \$m	0	0	75	190	302

a 70 bps rise in the yield curve. Thereafter they lose about \$2.3 million per basis point. On 2nd May 1994 the five-year and 30-year rates were 6.687% and 7.328% respectively, an average rate rise of over 150 bps.

In Figure 26.2 is shown the distribution of changes in US five-year rates over a six-month period during the 10 years prior to November 1993, data readily available at the time that the contract was signed.² This historical data suggests that there is a 14% chance of rates rising more than the 70 bps at which P&G start to lose out (the black bars in the figure). There is a 3% chance of a 150 bps or worse rise. Using this data to calculate the expected profit over the five-year period one finds that it is −\$8.7 million, rather than the hoped for +\$7.5 million.

When you get to the end of this chapter I want you to do a little exercise. Perform the above parallel shift calculation on a spreadsheet. Time yourself to find out how long it would have taken you to save \$200 million. (If you use Excel's built-in spreadsheet functions to calculate yields and prices then it should take less than 10 minutes of typing, and that includes switching on your PC and a comfort break.)


Figure 26.2 Distribution of changes in US five-year rates over a six-month period covering the 10 years prior to November 1993.

² I've cheated a bit in using overlapping data. Using non-overlapping data gives the same, or slightly worse for P&G, results.

The following, taken from the P&G website (www.pg.com), seems a decent enough principle (I don't know when it was written):

Integrity: We always try to do the right thing. We are honest and straightforward with each other. We operate within the letter and spirit of the law. We uphold the values and principles of P&G in every action and decision. We are data-based and intellectually honest in advocating proposals, including *recognizing risks*.

The italics are mine.

26.4 **METALLGESELLSCHAFT**

Metallgesellschaft (*pron.* Met Al gazelle shaft, emphasis on 'Al') is a large German conglomerate with a US subsidiary called MG Refining and Marketing (MGRM). In 1992 MGRM issued forward contracts to its clients, locking in the price of heating oil and gasoline for 10 years. The forward price was fixed at about \$3 above the then spot prices. Each month the client received a delivery of oil, paying fixed price. The contract also allowed for both parties to close the position. For example, the client could cancel the contract at any time that the shortest-dated oil futures price exceeded the fixed price. On exercise they would receive 50% of the difference in price between the short future and the fixed price, multiplied by the total volume of oil remaining on the contract. MGRM could also close some of the contracts if the short futures price exceeded some prescribed exit price.

These contracts proved popular with clients because they were the only long-dated contracts available with which to lock in a fixed price. No such contracts existed on an exchange. The total volume of oil in the contracts amounted to some 180 million barrels, the equivalent of 85 days of the output of Kuwait.

Since these contracts were OTC forwards no money changed hands until each delivery. At which time the net cash flow to MGRM was fixed minus spot. The lower the spot price the more that the contracts were of value to MGRM.

MGRM naturally wanted to hedge the oil price risk. But the only exchange traded contracts available were one- to three-month futures. MGRM implemented a strategy of hedging the long-term short OTC forward position with long positions in short-term futures traded on the New York Mercantile Exchange (NYMEX). Because the short-term contracts expired every few months the position had to be rolled over; as one position expired so another was entered into for the next shortest maturity. Theoretically this strategy would have been successful provided that MGRM had a good model for interest rates and the cost of carry. We will return to this point in a moment. Even if MGRM did have a decent model they hit problems because of the important distinction between futures and OTC forwards.

Oil prices fell during the latter half of 1993. Because they held long positions in the futures, a fall in price had to be met on a daily basis by an increase in margin. Futures are marked to market. The extra margin requirements amounted to \$900 million during 1993, a large sum of money. MGRM turned to its parent company to help with the funding. Metallgesellschaft responded by taking control of MGRM, installing a new management. In December 1993 the new management closed out half of the short-term contracts. Oil prices then started to rise in early 1994 so that MGRM started to lose out on the

long-term contracts. Their response was to cancel the OTC forward contracts and close all positions. By this time losses had amounted to \$1.3 billion.

But were they really losses? MGRM were losing out on the short-term futures positions as oil prices fell but remember that these contracts were for hedging purposes. As they lost money on the hedging position they also made money on the OTC forward contracts. The problem was that because these were forward contracts the profit on them was not realized until the positions expired. Marking to model should have resulted in a net flat position regardless of what the oil price did. Think back to margin hedging, discussed in Chapter 25.

Some say that the MGRM management panicked; they say otherwise.

26.4.1 Basis risk

There was a slight complication in this story due to the behavior of futures prices. The relationship between forward prices and spot prices is not as simple in the commodity markets as it is in the FX markets, for example. Arbitrage considerations lead to the theoretical result

$$F = Se^{(r+q)(T-t)}$$

where T is the maturity, r is the relevant risk-free yield and q is the cost of carry. This relationship leads to forward prices that are higher than spot prices, the graph of forward prices as a function of maturity would be upward sloping. In this case the market is said to be in **contango**.

In practice the strategy required to take advantage of the arbitrage opportunity is so impractical, involving the buying, transporting, storing, transporting, . . . of the commodity that the theoretical arbitrage is irrelevant. It is therefore possible, and even common, for the forward curve to be downward sloping. The market is said to be in **backwardation**.

In the Metallgesellschaft story the oil markets were in a state of backwardation initially. By rolling over the futures contract it was possible to make a profit, benefiting from the slope of the forward curve at the short end. During 1993 the spot oil price fell sharply and the market moved into contango. Now the rolling over led to losses, amounting to \$20 million per month. The question remains, did MGRM price into the OTC forward contracts this possible behavior of effectively the cost of carry? If they did (which seems unlikely, or even impossible *a priori*) then their hedging strategy would have been successful had it not been cut off in its prime.

It is difficult to build an accurate interest rate and/or cost of carry model. MGRM was therefore exposed to the risk of hedging one instrument with an imperfectly correlated one. Such a risk is generally termed **basis risk**.

26.5 GIBSON GREETINGS

I think we should use this as an opportunity. We should just call [the Gibson contact], and maybe chip away at the differential a little more. I mean we told him \$8.1 million when the real number was 14. So now if the real number is 16, we'll tell him that it is 11. You know, just slowly chip away at the differential between what it really is and what we're telling him.

... when there's a big move, you know, if the market backs up like this, and he is down another 1.3, we can tell him he is down another 2 ... If the market really rallies like crazy, and he's made back a couple of million dollars, you can say you have only made back a half a million.

February 23, 1994, BT Securities tape of a BT Securities manager discussing the BT internal valuation of the Gibson's positions and the valuation given to Gibson.

Gibson Greetings is a US manufacturer of greetings cards. In May 1991 they issued \$50 million worth of bonds with a coupon of 9.33% and with maturities of between four and 10 years. In the early 1990s interest rates fell and Gibson were left paying out a now relatively high rate of interest. To reduce the cost of their debt they entered into vanilla interest rate swaps with Bankers Trust in November 1991.

Swap 1: Notional \$30 million, two-year maturity, BT pay Gibson six-month LIBOR and Gibson pay BT 5.91%.

Swap 2: Notional \$30 million, five-year maturity, BT pay 7.12% and Gibson pay six-month LIBOR.

For the first two years the LIBOR cashflows cancel and Gibson receive $7.12 - 5.91 = 1.21\%$.

In July 1992 both swaps were canceled. Shortly afterwards Gibson entered into a more leveraged swap contract, and so began a sequence of buying and canceling increasingly complex products. Some of these products are described below.

Ratio swap: For five years Gibson were to pay

$$\frac{50}{3} r_L^2$$

to BT every six months and receive 5.5% on a notional of \$30 million. Here r_L is six-month LIBOR. Since the first fixing of LIBOR was at approximately 3.08% the first payment by Gibson was just 1.581%. The second payment was 1.893% corresponding to LIBOR of 3.4%.

Interest rates were starting to rise. And judging by the implied forward curve at the time the market was 'expecting' rates to go higher. Should six-month LIBOR reach $5.7\% (= \sqrt{0.06 \times 0.055})$ the net cashflow would be towards BT from Gibson. Fearing that rates would go beyond this level, the swap contract was amended three times and finally canceled in April 1993, six months after its initiation.

LIBOR did not rise as rapidly as the forward curve had suggested (it usually doesn't). With hindsight it probably would have been in Gibson's best interests to have retained the swap in its original form.

Periodic floor: For five years BT were to pay 28 bps above six-month LIBOR on a notional of \$30 million. Gibson were to pay

$$r_L$$

where r'_L is six-month LIBOR measured at the previous swap date as long as $r_L > r'_L - 0.15\%$. This is a path-dependent contract. At the end of 1992 BT informed Gibson that the value of the periodic floor was negative and it was canceled nine months after its initiation. The loss was incorporated into a later contract.

Treasury-linked swap: This swap was entered into in exchange for BT decreasing the maturity of the above ratio swap. Gibson were to pay BT LIBOR and BT were to pay Gibson LIBOR plus 2%. The catch was in the amount of the principal repayment at maturity. Gibson were to pay \$30 million while BT were to pay

$$\min \left(\$30.6 \text{ million}, \$30 \text{ million} \times \left(1 + \frac{103 \times Y_2}{4.88} - \frac{P_{30}}{100} \right) \right),$$

where Y_2 is the two-year Treasury yield and P_{30} is the 30-year Treasury price. Does this formula look familiar?

Knock-out call option: To reduce its exposure under the terms of the Treasury-linked swap due to an anticipated small principal repayment by BT, Gibson entered into a knock-out call option in June 1993. In return for an up-front premium Gibson were to receive at expiry

$$12.5 \times \$25 \text{ million} \times \max(0, 6.876\% - Y_{30})$$

where Y_{30} is the yield on 30-year Treasuries. The downside for Gibson was that the contract expired if the 30-year Treasury yield fell below 6.48%. Long-term rates would have had to stay remarkably stable since the payoff would only be received if $6.48\% < Y_{30} < 6.876\%$ at expiry with Y_{30} *never* having fallen below 6.48%.

The time swap/corridor swap: In August 1993 Gibson entered into a corridor swap with BT. The maturity of the contract was three years and the notional \$30 million. BT were to pay Gibson six-month LIBOR plus 1%. Gibson's payments were as follows:

$$r_L + 0.05 \times N\%$$

where r_L is six-month LIBOR and N is the number of days during each six-month calculation period that r_L was outside the range specified in Table 26.2.

After much modification of the multiplier (0.05 initially), the ranges and the termination date, the contract was canceled in January 1994.

In all there were approximately 29 contracts. Gibson's losses amounted to over \$20 million, equivalent to almost a year's profits. They sued BT and in an out-of-court settlement agreed in November 1994 they were released from paying BT \$14 million outstanding from some swap contracts, only paying BT \$6.2 million. Gibson's argument was that they had been misled by BT as to the true value of their contracts. As the tape recording transcribed at the start of this section shows, BT had their own internal models for the value of the contracts but consistently understated the losses to Gibson.

Table 26.2 Ranges for LIBOR in the corridor swap.

6 Aug 1993–6 Feb 1994	3.1875–4.3125%
6 Feb 1994–6 Aug 1994	3.2500–4.5000%
6 Aug 1994–6 Feb 1995	3.3750–5.1250%
6 Feb 1995–6 Aug 1995	3.5000–5.2500%

26.6 BARINGS

The Barings story is actually very simple, and the role of derivatives is relatively small. The reasons for all the fuss are the magnitude of the losses (\$1.3 billion), that it involved a very staid, 200-year old bank, and that its main protagonist, 28-year old Nick Leeson, did a runner.

Nick Leeson was a trader for Barings Futures, working in Singapore. In 1994 he was reported to have made \$20 million for Barings and was expecting to be rewarded with a bonus of \$1 million on top of his \$150,000 basic. It is likely that he had in fact lost a considerable amount already, with the real position only appearing in the infamous 'Error Account 88888.' Leeson had been given too much freedom. At the time, late 1994 and early 1995, Leeson had control over both a trading desk and back office operations; in effect he was allowed to police his own activities.

The trades that caused the downfall of Barings involved the Japanese stock market, specifically futures and options on the Nikkei 225 index. In November and December 1994 Leeson had been selling straddles on the Nikkei with strikes in the region of 18,500–20,000. The Nikkei index was then trading in a similar range. As long as the index was stable and volatility remained low he would profit from this almost delta-neutral strategy. On 17th January 1995 an earthquake hit Kobe and the Nikkei started to fall, see Figure 26.3. The figure also shows Leeson's trading range. Over the next few days Leeson began buying index futures with March expiry. One of the reasons behind this strategy was the belief that the magnitude of his trades would act to reverse the declining market. If he could bring the Nikkei up to the pre-earthquake levels his option positions would be safe.

Although his trades had a significant impact on the index they could not hold back the fall. Over the next month the index was to fall to 17,400. As the index continued its fall, Leeson increased his trades to shore up the index. Margin was required for the futures mark-to-market and vast sums of money were transferred from Barings in London. Finally,

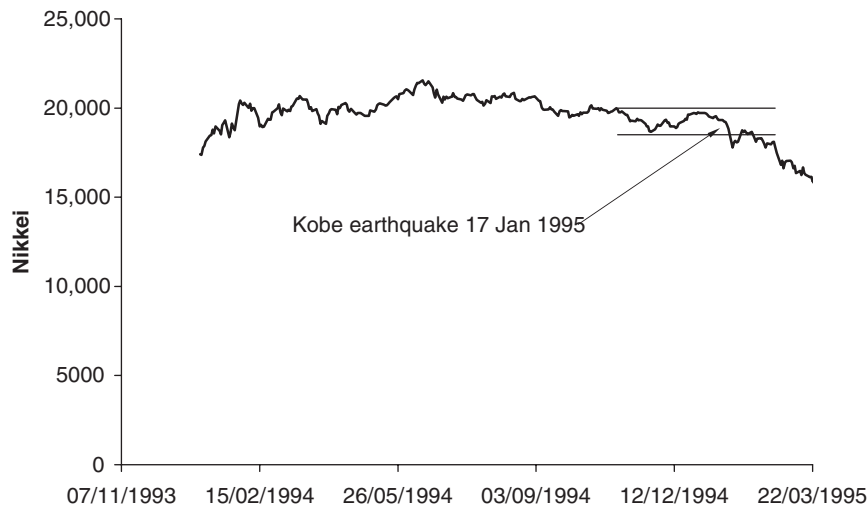


Figure 26.3 The Nikkei 225 index from the beginning of 1994 until March 1995.

the margin calls became too much to cover. On 23rd February 1995 Nick Leeson went on the lam, fleeing Singapore. Finally, he took a flight to Frankfurt, where he was arrested. On 1st December 1995 he pleaded guilty to two offenses of deceiving Barings' auditors in a way likely to cause harm to their reputation as well as cheating SIMEX. The next day he was sentenced to six and a half years in prison.

Leeson was released from the Tanah Merah wing of Changi jail on 3rd July 1999. While in prison he developed cancer of the colon, his wife divorced him and remarried, and they made a movie of the story starring Ewan McGregor and Anna Friel.

Meanwhile Barings went broke. The Sultan of Brunei was approached to bail them out but declined. The Dutch bank ING 'was finally persuaded to take on the corpse of Barings,' as Richard Thomson put it, for the grand sum of £1.

26.7 **LONG-TERM CAPITAL MANAGEMENT**

Long-Term Capital Management (LTCM) is a hedge fund. Hedge funds are supposed to hedge, you'd think. Yet there are few regulations governing their activities. The term 'hedge fund' came about because these funds take short as well as long positions, but this is *not* the same as hedging. LTCM was founded by John Meriwether, ex-Salomon bond arbitrage team, with Nobel Laureates Myron Scholes and Robert Merton as partners in the firm. (The excellent book by Lewis is an account of dodgy dealings at Salomon's during Meriwether's time there.)

Edward Thorp (of blackjack fame, see Chapter 20) started one of the first hedge funds in 1969. It has been a very successful hedge fund and Thorp was invited to invest in LTCM in 1994 when it was founded. Thorp declined, 'I didn't want to have anything to do with it because I knew these guys were just dice rollers. It was just a mutual admiration society at Long-Term and nobody was focusing clearly enough on the model.'

If Thorp didn't want to invest, he was in the minority. It was a glamorous line-up and many who should have known better got sucked in. For the first couple of years they made good returns, of the order of 40%. But look at Figure 26.4, *everyone* was making 40%. And in 1997 they made 27%.

The hype surrounding LTCM, its legendary team and the friends they had in high places meant that they had three benefits that other firms would kill to have:

- They were able to leverage \$4.8 billion into \$100 billion. Its notional position in swaps was at one time \$1.25 trillion, 5% of the entire market.
- They were excused collateral on many deals.
- When they finally went under the Federal Reserve organized a bail-out.

During 1998 LTCM took huge leveraged bets on the relative value of certain instruments. They were expecting a period of financial calm and convergence of first and emerging world interest rates and credit risk. Here are some of the strategies LTCM had in place, what they gambled would happen and what actually did happen. Observe how many of these trades come in pairs; LTCM were making relative value trades.

European government bonds LTCM sold German Bunds and bought other European bonds in the expectation of interest rate convergence in the run-up to EMU. They did ultimately converge but not before first diverging further.



Figure 26.4 The S&P500.

Emerging market bonds and US Treasuries LTCM had long positions in Brazilian and Argentinian bonds and short positions in US Treasuries. They expected credit spreads to decrease, instead they widened to as much as 2000 bps.

Russian GKO's and Japanese bonds They expected Russian yields to fall and Japanese to rise, and so bought Russian GKO's and sold Japanese bonds. They were wrong about the direction of Japanese yields, and Russia defaulted.

Long- and short-term German bonds LTCM bought 30-year German bonds, sold 10-year bonds, expecting the German yield curve to flatten. Instead, demand for the short-term bonds caused the yield curve to steepen.

Long- and short-dated swaption straddles They bought long-dated swaption straddles and sold short-dated swaption straddles. This is an almost delta-neutral strategy and therefore a volatility trade. Short-term volatility rose, resulting in losses.

The main point to note about these trades is that all seem to have the same view of the world market. They were expecting a period of relative stability, with emerging markets in particular benefiting. In fact, the default by Russia on 17th August 1998 sent markets into a panic, investors 'fled to quality' and all the above trades went wrong. LTCM had most definitely made a big bet on one aspect of the market, and were far from being diversified. Along the way the partners lost 90% of their investment and a couple of Nobel prize winners had very red faces. They say on Wall Street that if you lose \$5 million you'll never work on the Street again, lose \$50 million and you'll walk into a new job the next day. But lose billions? It's good to have friends.

Figure 26.5 shows the price of the $9\frac{1}{4}\%$ 2001 USD Argentine bond. Prior to the August Russian default the price is quite stable. But then the price plummeted in the overall panic. As things calmed down the price returned. LTCM may have been correct in their views

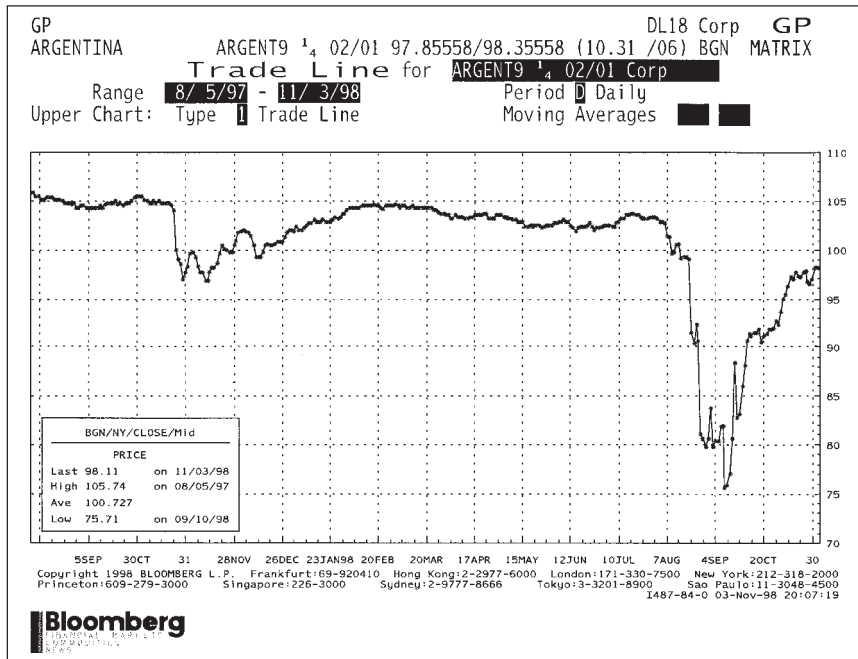


Figure 26.5 Time series of the 9 $\frac{1}{4}$ % 2001 USD Argentine bond. Source: Bloomberg L.P.

long term, but if you can't weather the storm in the *short term* your correct market view is irrelevant. As hedge fund is a misnomer, so, perhaps, is LTCM... a more accurate name might be STCMM.

On 21st August 1998 they managed to lose \$550 million, and there was more to come.

It was deemed too dangerous for LTCM to be allowed to fail completely, the impact on the US economy could have been disastrous. So the New York Federal Reserve organized, in September 1998, a bail-out in which 14 banks invested a total of \$3.6 billion in return for a 90% stake in LTCM.

Much of the LTCM story is still unknown, they were so glamorous that many of their investors didn't even know what they were up to. However, it seems that LTCM were using only rudimentary VaR estimates, with little or no emphasis on stress testing. They estimated that daily swings in the portfolio should be of the order of \$45 million. But this didn't allow for extreme market moves and problems with liquidity. In trying to reduce risks LTCM even sold off liquid assets, leaving the (theoretically) more profitable illiquid trades stay on their books. Not a sensible move at times of crisis. Simple VaR as described in Chapter 22 is fine as far as it goes, but cannot deal with global financial meltdowns.

'When we examine banks, we expect them to have systems in place that take account of outsized market moves,' said Alan Greenspan, chairman of the Federal Reserve.

There's another aspect to the LTCM tale that should make anyone using simple quantitative analysis a little bit wary. And that is the matter of liquidity. In August 1998 there was a worldwide drying up of liquidity. Without liquidity it is impossible to offload your positions and the idea of 'value' for any product becomes meaningless. You just have to wait until the market decides to loosen up. Part of the problem was that many

of the banks they dealt with knew of LTCM's trades. Many of these were simply copying LTCM's strategies. This wouldn't have mattered if LTCM and all these banks were dealing in small quantities. However, they weren't and in some cases completely cornered the market. It's one thing to corner the market in one of the necessities of life, such as beer, but to corner the market in something obscure makes you a sitting duck when you want to sell.

26.8 **SUMMARY**

Most of these stories have similar themes, overconfidence, lack of understanding of the risks, pure speculation at inappropriate times or for inappropriate reasons, overgearing.

We're only human but what is the point of the math modeling when any profits gets thrown away by a few individuals who don't know what they are doing? If they *do* know what they are doing, I would assume they are crooks.

I'll end with a note of caution sounded by Robert Merton in 1993: 'Any virtue can become a vice if taken to extreme, and just so with the application of mathematical models in finance practice . . . At times the mathematics of the models become too interesting and we lose sight of the models' ultimate purpose. The mathematics of the models are precise, but the models are not, being only approximations to the complex, real world . . . The practitioner should therefore apply the models only tentatively, assessing their limitations carefully in each application.' Doh!

FURTHER READING

- Leeson (1997) explains the Barings disaster from his own perspective.
- Chew (1996) is an excellent book covering the risks of derivative transactions. She discusses the technical side of contracts, the legal side and the morality.
- Partnoy (1998) is a cracking good tale of goings-on at Morgan Stanley (coincidentally one of my least favorite banks).
- For background on Meriwether, the LTCM partner, see Lewis (1989).
- Dunbar (1998), Kolman (1999) and Jorion (1999) are nice accounts of LTCM.
- Miller (1997) discusses Metallgesellschaft in detail.
- Thomson (1998) has plenty of inside gen on many of the derivatives stories.
- Read Merton (1995) for his thoughts on the influence of mathematical models on the finance world.